

Numerical Analysis I, Midterm Exam I, 27 November 2020

1) Use the bisection method and perform three iterations to find solution for $x - 0.8 - 0.2 \sin x = 0$ on $[0, 1]$. How many iterations of the bisection method are required to estimate this root an accuracy of 10^{-4} .

2) a) Show that $g(x) = \sqrt{2 - \cos x}$ has a fixed point on $[-1, 2]$.

b) Use fixed point iteration method to determine a solution accurate within 10^{-3} for $xe^x - 1 = 0$.

3)

x	1	2	3	5
$y = f(x)$	1.732	2.236	2.645	3.316

Use inverse interpolation to find the value of x for $y = 3$ constructing **divided differences** table for the data given above.

Note: in all calculations, use three digits after the decimal point.

Duration: 80 min.